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E-250 UNIT METERING AND PROTECTION PANEL

1. SCOPE OF WORK

The Work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of the unit metering and protection equipment in the combined cycle power plant in accordance with Part 4, Subsection 1.2 Summary of Work.

The following equipment shall be provided with steam turbine generator units:

- (1) Generator protection panel.
- (2) Generator Transformer protection panel.
- (3) Unit Auxiliary Transformer (UAT) protection panel.
- (4) Unit tariff metering instrumentation panel.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards:

2.1 National

- | | | |
|------|---|--|
| PUIL | : | Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation – Indonesia). |
| SPLN | : | Standard Perusahaan Umum Listrik Negara. |

2.2 International Electrotechnical Commission

- | | | |
|-----------|---|--|
| IEC 60051 | : | Direct acting indicating analogue electrical measuring instruments and their accessories. |
| IEC 60145 | : | Var-hour (reactive energy) meters |
| IEC 60255 | : | Electrical relays |
| IEC 61010 | : | Safety requirements for electrical equipment for measurement, control, and laboratory use. |
| IEC 61554 | : | Panel mounted equipment-Electrical measuring instruments-Dimensions for panel mounting. |

IEC 62052	:	Electricity metering equipment (AC) General requirements, tests and test conditions.
IEC 62053	:	Electricity metering equipment (AC) Particular requirements.

3. OPERATING AND DESIGN CONSIDERATIONS

The unit metering and protection panel shall be designed for operation in a combined cycle power plant located in a tropical climate along seacoast. Refer to Part 4, Subsections 1.1, 2.14, and 2.16 for the meteorological, seismic and environmental condition at the site.

Particular note should be taken of the fact that even though the protection panels will be located in an air-conditioned area, they must be able to function normally if the air conditioning system fails.

4. DETAILED REQUIREMENTS

4.1 General

The metering and protection panels shall be free standing enclosures. The construction of the panels shall be in accordance with Technical Specification E-280. The panels layout shall be subject to the approval of the Owner.

Indicating instruments shall be of accuracy class 0.5.

All Watt hour and Var hour meters shall have pulse initiators suitable for use by the Owner. They shall be of accuracy class 0.2.

Transducers and miscellaneous electrical devices shall be in accordance with the applicable requirements of Technical Specification E-001.

The metering and protective functions provided shall, as a minimum, be as shown on drawings. The Contractor may suggest additional metering and protective functions to the Owner. The design responsibility for adequately protecting the equipment shall be the Contractor's.

The Contractor shall undertake the detailed design of tripping schemes and coordination of relay settings throughout the equipment supplied by him, in line with the general concept given in the specification drawings and this specification. A complete relay coordination study shall be submitted for the

approval of the Owner, to ensure compliance with power system protection, prior to final selection of relay equipment.

The Contractor shall be responsible for calculating the transducer constants and the settings of all relays. Setting calculations and setting tables shall be submitted for the review of the Owner as part of the above coordination study.

4.2 Relays

The relays shall be in accordance with technical specification E-001. The protective relays shall be digital multifunction duplicated microprocessor type with self inspection function.

4.3 Generator Protection

This section describes the minimum requirement of protective relays required for the generator protection.

A high speed percentage restraint differential relay, 87G, shall be provided. Harmonic restraint is not required. The relay shall be designed for differential protection of rotating generators and shall have a percent bias slope of about 10%. Operation of this relay shall trip the turbine-generator.

A high speed overall percent differential relay, 87GT, shall be provided. The relay shall have harmonic restraint to avoid false operation due to inrush current to the generator transformer and unit auxiliary transformer. The relay shall trip the turbine generator.

A time over-voltage relay, 59GA, shall be provided to monitor the voltage across the generator neutral resistor to detect ground faults in the generator and the high voltage windings of the unit auxiliary transformers. The relay shall respond to 50 Hz voltages only and shall be sensitive enough to protect 90% or more of the generator stator winding. The relay shall also be able to withstand the voltage appearing across the neutral resistor during a ground fault. The relay shall have a time overcurrent characteristics. The relay shall be set to coordinate with the voltage transformer fuses so that a fault on the voltage transformer secondary will not cause relay operation. The relay shall trip the turbine generator.

A second time over-voltage relay, 59BUG, shall be connected across broken delta connected auxiliary voltage transformer. The relay shall have harmonic filtering as described in the previous 59GA. The relay shall be able to detect faults in 90% of the generator stator winding and the high voltage windings of the generator transformer and unit auxiliary transformers. Operation of the relay shall be supervised by the voltage balance relay 60G, to prevent operation due to the opening of a voltage transformer fuse. The relay shall trip the turbine generator.

A third stator ground relay, 59GB, shall be provided to detect 100% ground faults over the entire stator winding. The relay shall give alarm only.

A loss of field relay 40G, shall be provided to detect the loss of excitation to the generator. The relay shall be of the distance type with any necessary directional or blinder elements. The diameter of the impedance circle and the circle offset shall be independently adjustable. If a timer is needed to avoid operation during stable power swings, it shall be provided. If the openings of a voltage transformer fuse could cause false tripping of the relay, it shall be supervised by voltage balance relay, 60G1. The relay shall be set to avoid false tripping during power swings where it is possible for the machine to remain stable. The relay shall trip the turbine generator.

The Contractor shall provide dual level volts per hertz protection (relay 59/81 G1-1 and 59/81 G1-2). The relays shall protect the generator and the transformers connected to the generator from over excitation. The relays shall detect the ratio of voltage to frequency. The first relay (59/81 G1-1) shall pick up at about 1.1 per unit volts per hertz. It shall alarm and then trip the turbine generator after about 45 seconds, the second relay (59/81 G1-2) shall pick up at about 1.2 per unit volts per hertz and shall trip the turbine generator after about six (6) seconds. The Contractor shall determine the final settings for the relays. The settings of the volts per hertz relays shall be coordinated with the excitation limiter provided as part of the voltage regulator.

The Contractor shall provide dual anti-motoring relays. These relays (32AG and 32BG) shall detect inadequate steam flow to the steam turbine and fuel flow to gas turbine by detecting reverse power flow to the prime mover. The relays shall have the ability to detect 0.5% reverse power. The effects of the instrument transformers on the sensitivity of the reverse power relay shall be

considered in the design of the system. The relays shall alarm on pick up. The relays shall trip the turbine generator when the trip contacts of either relay close.

The Contractor shall provide out-of-step protection by means of a combination of relays, 121 G. This shall consist of a combination of an offset mho relay, blinder relays and a counter arranged to detect the shift in apparent impedance at the relay location associated with loss of stability of the unit. The relay system shall be arranged to disconnect the generator from the system and operate the turbine bypass system when the generator loses stability.

The Contractor shall provide a negative sequence current time-overcurrent relay, 46G, to provide protection against sustained unbalanced faults or severely unbalanced loads appearing at the generator terminals. The characteristics and setting of this relay shall be coordinated with the negative sequence capability of the generator, and it shall provide protection from all negative sequence current conditions that might damage the machine. Operation of this relay shall cause the generator to be disconnected from the extra high voltage system and the turbine bypass to be activated.

The Contractor shall provide a backup impedance relay, 21G, that will detect faults both in the generator and in the extra high voltage system. The relay shall be equipped with a timer that will allow the backup substation relays to clear system faults before the relay operates. The relay shall be supervised by the voltage balance relay, 60G, to prevent false operation in case of a fault in the voltage transformer circuit. Operation of the relay shall cause the generator to be disconnected, from the extra high voltage system and the turbine bypass to be activated.

The Contractor shall provide two (2) voltage balance relays, 60G, to supervise the operation of all generator protective relays that may trip false if they lose their potential source. The relays shall compare the voltage appearing in the secondary of each to the two (2) sets of generator voltage transformers. The relay shall operate if the difference in voltage between the circuits exceeds a certain value. Operation of the relays shall disable all relays that may trip due to the loss of potential signal.

The Contractor shall provide under frequency protection (81G) to protect the turbine against damage resulting from operation of the turbo-generator at below

rated frequency. The under-frequency trip relay shall only be enabled when the generator has been connected to the system. Operation of this relay shall trip the turbine generator.

4.4 Generator Transformer Protection

The Contractor shall provide a percent differential relay, 87GT, for protection against internal faults in the generator transformer. The relay shall be equipped with harmonic restraint to prevent false tripping due to transformer magnetizing current. The relay shall be of the high speed which shall be able to operate within one (1) cycle of the inception of a severe internal fault. The operation of the relay shall cause the turbine generator to trip.

The Contractor shall provide a ground relay for the neutral of the high voltage winding of the generator transformer, 51 GGT. The relay shall be selected and set so that it coordinates with substation relays. Operation of the relay shall cause the turbine generator to trip.

The Contractor shall provide all necessary auxiliary relays and tripping circuits required for the sudden pressure relays, 63GT, at the transformer. Operation of this relay shall cause the turbine generator to trip.

4.5 Unit Auxiliary Transformer (UAT) Protection

The Contractor shall provide independent but identical protection systems for each unit auxiliary transformer.

The Contractor shall provide harmonic restraint percentage differential relays, to detect internal faults in the UAT. Operation of the relay shall cause the turbine generator to trip.

Each UAT shall be provided with a primary overcurrent relays. Each relay shall consist of an instantaneous overcurrent element and an inverse or a definite time-delay overcurrent relay. The relay shall be able to provide back up protection against internal transformer faults on the connections to the gas turbine and steam turbine 6 kV unit buses. The relay type and setting shall be chosen so that the relay operation is coordinated with the downstream relays. Operation of these relays shall cause the turbine generator to trip.

The Contractor shall provide inverse or definite time ground overcurrent relays to detect internal and uncleared external ground faults in the low voltage windings of the UAT. The relay type and setting shall be chosen so that the relay operation will be coordinated with the downstream relays. Operation of the relays shall cause the turbine generator to trip.

The Contractor shall provide all necessary auxiliary relays and other accessories necessary to include the sudden pressure relays provided by the high voltage transformer in the protection scheme. Operation of a sudden pressure relay shall cause the turbine generator to trip.

4.6 Synchronizing

4.6.1 Generator

The Contractor shall provide a package for synchronizing each steam turbine generator. Separate relays and instruments shall be provided for each generator.

The generator synchronizing equipment shall allow either manual or automatic synchronizing of the machine. In the latter mode, the DCS for gas turbine unit or steam turbine unit shall adjust the turbine governor and field to bring the generator into synchronism with the system and shall determine the exact moment for closing the breaker.

Separate synchronizing equipment shall be provided for each unit. The manual instruments shall include a synchroscope, frequency meter, incoming and running source voltmeters. One (1) common manual synchronizing equipment for GTG and STG shall be provided in each central control room.

Both automatic and manual synchronizing shall be supervised by a separate synchrocheck relay. This relay shall block an attempt to close the circuit breaker when the phase difference between the incoming and running systems are greater than ± 10 degrees. If an attempt to close the breaker is blocked by this relay, it shall be necessary for the operator to release the circuit breaker control switch before another attempt to synchronize is made.

4.7 Auxiliary Power and Tariff Metering

The Contractor shall be responsible for the detailed design of the auxiliary power and tariff metering system in accordance with the requirements outlined herein. The Contractor shall provide all required current, voltage, power and

other electrical signals for metering. The indicating instruments shall be installed on the metering panel.

The metering system shall measure the power generation at the generator terminals and the power consumption of the unit auxiliaries at the high voltage winding of the unit auxiliary transformer. The net generation (export) and power supply for start up and during shutdown (import) shall also be measured at the high voltage side of the generator transformer using a voltage and current signals.

5. PERFORMANCE TESTING AND GUARANTEES

The performance testing and guarantees for unit metering and protection panel and equipment shall be as described in Technical Specification E-280 except as modified/supplemented below:

Type tests shall be performed on all equipment as required by the applicable industry standards. Type test report and/or certificate of them will be acceptable.

All assemblies shall be subjected to a high voltage test of 1500 volts to ground for one (1) minute. Electronic assemblies that are not designed to withstand such a test shall be disconnected while the test is performed.

The correct wiring and correct functioning of the panels shall be verified before shipment to the site.

Relay settings shall be proved by appropriate primary and secondary current injection tests in the field.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for all unit metering and protection equipment shall conform to the requirements included in Part 4, Subsection 2.7.

The quality assurance/quality control procedures specified shall include, but not be limited to, the followings:

- (1) Design criteria and engineering calculations
- (2) Environmental tests
- (3) Continuity and wiring checking.

- (4) High voltage tests
- (5) Instrument calibration
- (6) Relay settings, calibration and tests
- (7) Control and functional testing.
- (8) All routine tests and special tests specified
- (9) All proposed site tests to verify the unit metering and protection system performance

The quality assurance/quality control documentation as specified above, shall include, but not be limited to, the followings:

- (1) Inspection Reports
- (2) Certified Test Reports
- (3) Reports of routine tests and any other special tests performed at the factory
- (4) Reports of all the site tests to verify the unit metering and protection system performance
- (5) Relay setting calculations
- (6) All other documentation required by applicable codes and standards

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